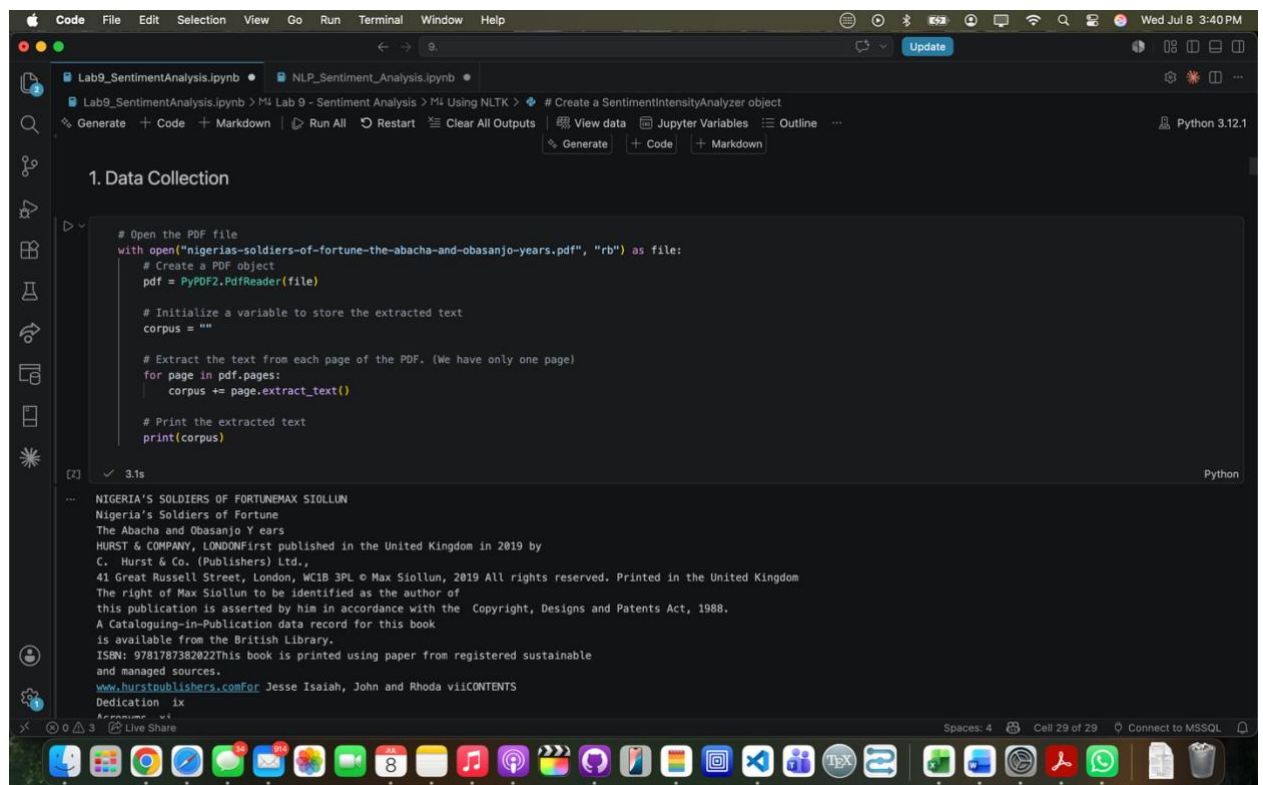


Lab Instructions:

The goal of this lab is to perform sentiment score classification based on a PDF document. Perform the following steps to complete the assignment. Make sure to attach relevant screenshot after each step.

- Download a PDF copy of any ebook (Choose one of your favourite books – not a textbook). Modify the notebook (NLP_Sentiment_Analysis) to Open the PDF copy of ebook.

Place a screen shoot of modified code here.



The screenshot shows a Jupyter Notebook interface with the following content:

```
# Open the PDF file
with open("nigerias-soldiers-of-fortune-the-abacha-and-obasanjo-years.pdf", "rb") as file:
    # Create a PDF object
    pdf = PyPDF2.PdfReader(file)

    # Initialize a variable to store the extracted text
    corpus = ""

    # Extract the text from each page of the PDF. (We have only one page)
    for page in pdf.pages:
        corpus += page.extract_text()

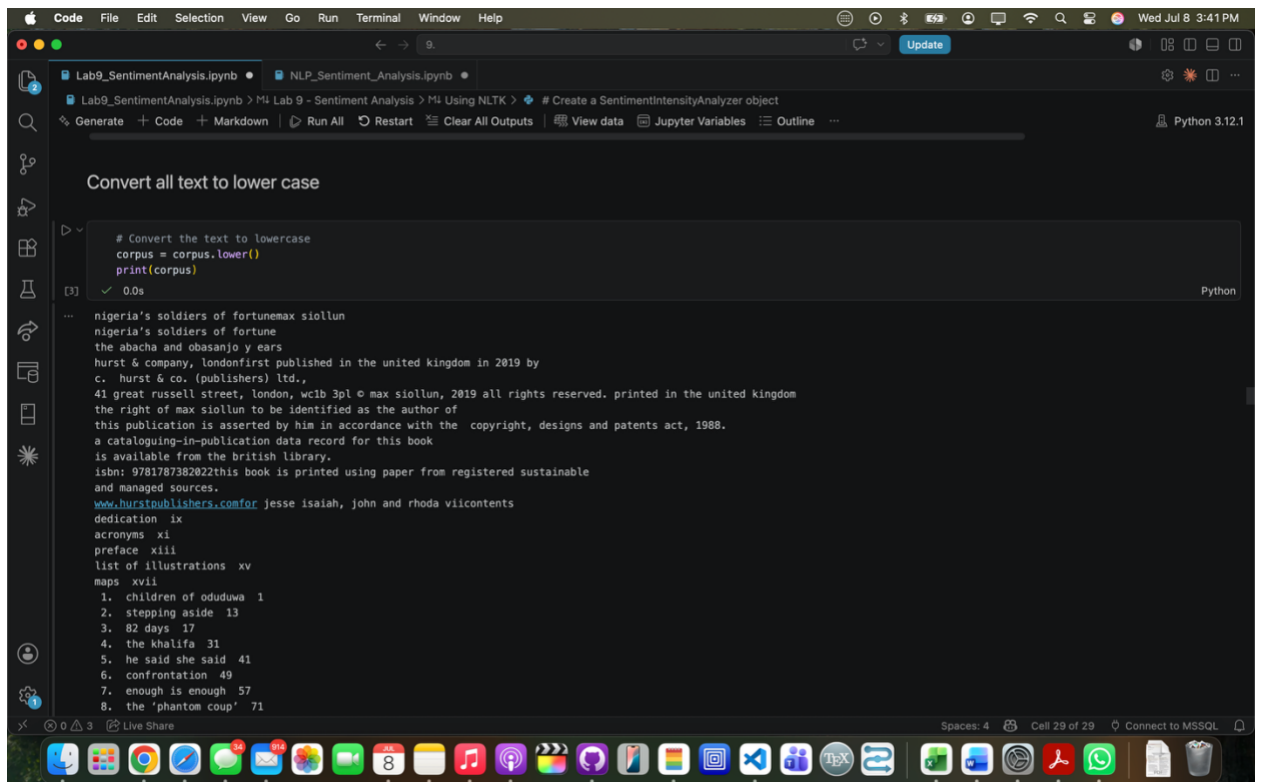
    # Print the extracted text
    print(corpus)
```

The output of the code is displayed below the cell:

```
... NIGERIA'S SOLDIERS OF FORTUNE MAX SIOLLUN
Nigeria's Soldiers of Fortune
The Abacha and Obasanjo Years
HURST & COMPANY, LONDON First published in the United Kingdom in 2019 by
C. Hurst & Co. (Publishers) Ltd.,
41 Great Russell Street, London, WC1B 3PL © Max Siollun, 2019 All rights reserved. Printed in the United Kingdom
The right of Max Siollun to be identified as the author of
this publication is asserted by him in accordance with the Copyright, Designs and Patents Act, 1988.
A Cataloguing-in-Publication data record for this book
is available from the British Library.
ISBN: 9781787382822 This book is printed using paper from registered sustainable
and managed sources.
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```

- Apply all the preprocessing steps (text to lowercase, tokenization, punctuation, stemming, lemmatization, stop words, spell checks). Make sure to use at least 5 acronyms from the pdf.

Place relevant screen shoots of all modified code here.

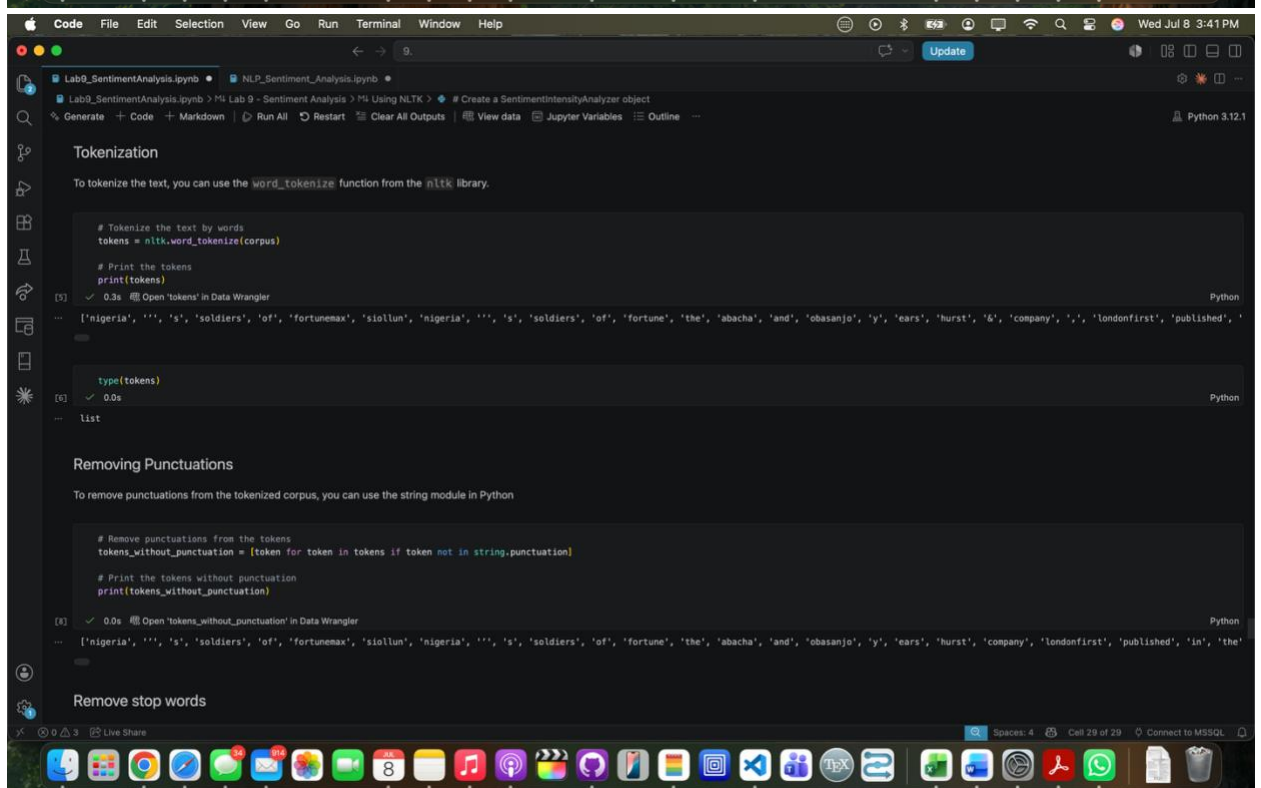


The screenshot shows a Jupyter Notebook in VS Code with the following content:

```
# Convert the text to lowercase
corpus = corpus.lower()
print(corpus)
```

The output of the cell is:

```
nigeria's soldiers of fortunemax siollun
nigeria's soldiers of fortune
the abacha and obasanjo y ears
hurst & company, londonfirst published in the united kingdom in 2019 by
c. hurst & co. (publishers) ltd.,
41 great russell street, london, wc1b 3pl © max siollun, 2019 all rights reserved. printed in the united kingdom
the right of max siollun to be identified as the author of
this publication is asserted by him in accordance with the copyright, designs and patents act, 1988.
a cataloguing-in-publication data record for this book
is available from the british library.
isbn: 9781787382022this book is printed using paper from registered sustainable
and managed sources.
www.hurstpublishers.comfor jesse isalah, john and rhoda viicontents
dedication ix
acronyms xi
preface xiii
list of illustrations xv
maps xvii
1. children of oduduwa 1
2. stepping aside 13
3. 82 days 17
4. the khalifa 31
5. he said she said 41
6. confrontation 49
7. enough is enough 57
8. the 'phantom coup' 71
```



The screenshot shows a Jupyter Notebook in VS Code with the following content:

```
# Tokenize the text by words
tokens = nltk.word_tokenize(corpus)

# Print the tokens
print(tokens)
```

The output of the cell is:

```
['nigeria', "'", 's', 'soldiers', 'of', 'fortunemax', 'siollun', 'nigeria', "'", 's', 'soldiers', 'of', 'fortune', 'the', 'abacha', 'and', 'obasanjo', 'y', 'ears', 'hurst', '&', 'company', ',', 'londonfirst', 'published', 'in', 'the', 'phantom coup', '']
```

The next cell shows the type of the tokens:

```
type(tokens)
```

The output is:

```
list
```

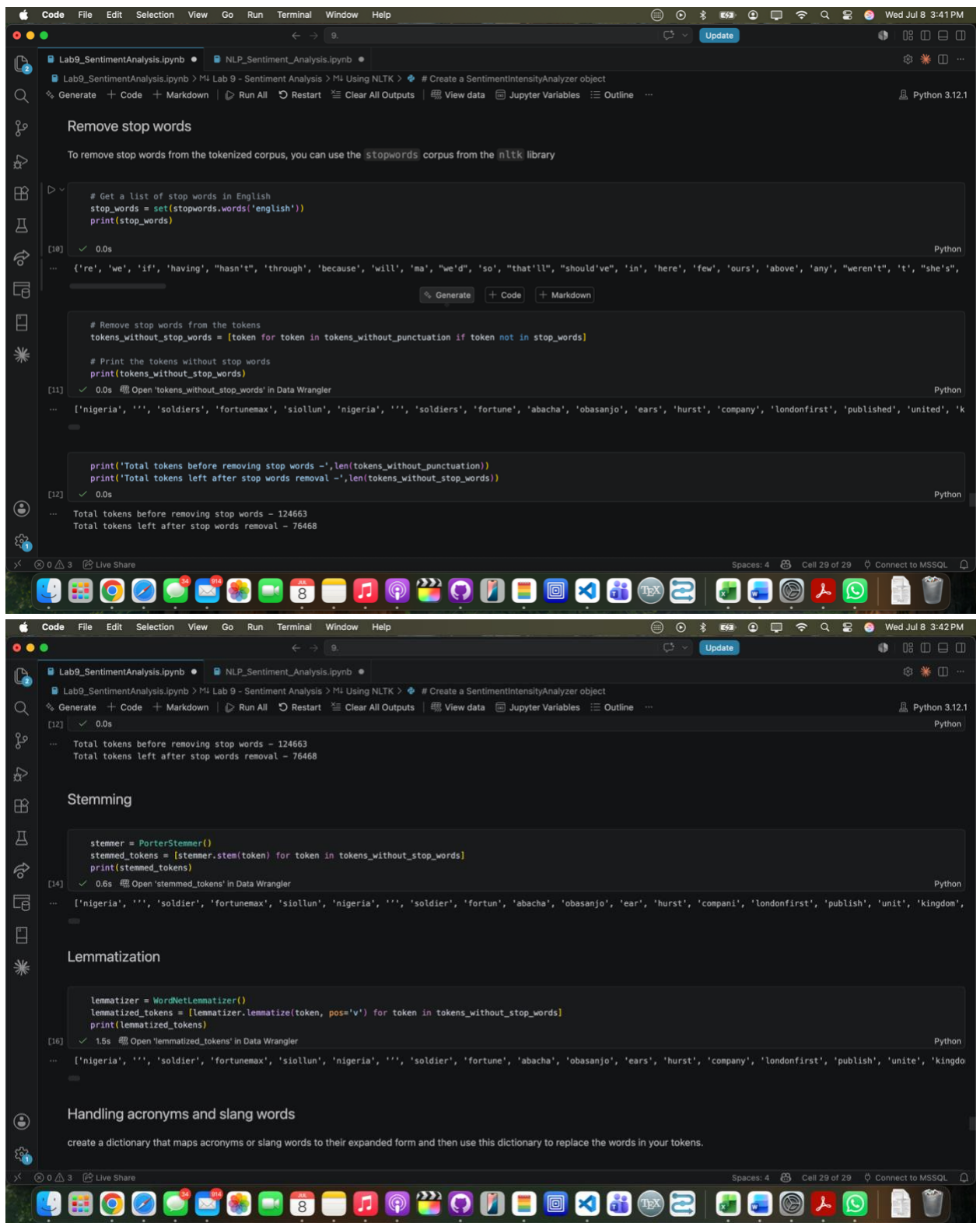
The next section is titled "Removing Punctuations". The code in the cell is:

```
# Remove punctuations from the tokens
tokens_without_punctuation = [token for token in tokens if token not in string.punctuation]

# Print the tokens without punctuation
print(tokens_without_punctuation)
```

The output of the cell is:

```
['nigeria', "'", 's', 'soldiers', 'of', 'fortunemax', 'siollun', 'nigeria', "'", 's', 'soldiers', 'of', 'fortune', 'the', 'abacha', 'and', 'obasanjo', 'y', 'ears', 'hurst', 'company', 'londonfirst', 'published', 'in', 'the', 'phantom coup', '']
```



- Use two different scenarios one with stemming and one with lemmatization to calculate the sentiment score with both NLTK and textblob.

Place relevant screen shots of modified code and outputs.

The image displays two screenshots of a Jupyter Notebook titled "Lab9_SentimentAnalysis.ipynb" in a VS Code environment. The notebook is using Python 3.12.1.

Top Screenshot:

- Section: Fixing Typos**
- Code:** A list comprehension uses `spell.autocorrect_word(token)` to correct tokens. It prints the original and corrected tokens.
- Output:** Shows the original tokens (including typos like 'nigeria', 'soldier', 'fortunemax', 'siollun') and the corrected tokens (e.g., 'nigeria', 'a', 'soldier', 'fortunemax', 'swollen').
- Section: Using TextBlob**
- Code:** Joins the corrected tokens into a string and uses `TextBlob(text)` to analyze sentiment.
- Output:** Prints the sentiment analysis result: `Sentiment(polarity=0.83584286225362077, subjectivity=0.35222787875919875)`.

Bottom Screenshot:

- Section: Using NLTK**
- Code:** Creates a `SentimentIntensityAnalyzer` object, calculates sentiment scores for each token, and then calculates an average compound sentiment score. It includes a logic block to classify the sentiment as "positive", "negative", or "neutral" based on the score.
- Output:** Prints the sentiment score and the classified sentiment: `Sentiment Score: 0.8022581341214625724` and `Sentiment: neutral`.

- Summarize the results.

The pdf is a book about Nigeria during the military regime of Abacha. With a polarity score of 0.035 and subjectivity of 0.35, this is a neutral.

A sentiment score of 0.0022, this book has a neutral sentiment.

This Assignment will be demoed on July 15, Wednesday.